

Management of Herpes Simplex Virus (HSV) in People Living with HIV

Resistance to Anti-Herpes Drugs

Resistance to anti-herpes drugs is more common in individuals with HIV co-infection and is associated with treatment failure for genital herpes. Suppressive antiviral therapy with currently available agents has been shown in multiple studies to have no impact on HIV acquisition or transmission risk. Therefore, HSV treatment should not be used solely to manage or reduce HIV transmission or acquisition risk.

Much of the evidence on herpes management in people living with HIV comes from studies conducted before the widespread use of combination antiretroviral therapy (ART). Early prospective studies demonstrated that clinical lesions could be persistent and progressive in this population. Genital herpes, including chronic erosive lesions, may also occur as a manifestation of immune reconstitution inflammatory syndrome (IRIS) following the initiation of ART.

Patients with low CD4 counts are typically more susceptible to HSV-2 shedding, with a high risk of lesion development within the first 6 months after starting ART—particularly during the first 2 months. High levels of HSV viral shedding are associated with elevated HIV viral load prior to ART initiation. HSV-associated IRIS may be unresponsive to previously effective anti-herpes therapy, even in the absence of documented antiviral resistance. Management can be challenging; however, judicious use of prednisolone, as used in other forms of IRIS, may be beneficial.

Management of Initial Episode HSV in People Living with HIV

There are no clinical trial data supporting the use of any antiviral for first-episode genital HSV specifically in people living with HIV. Most adults with HIV have serological evidence of prior HSV-1 and HSV-2 infection, making acquisition studies difficult to conduct.

Case reports suggest that primary genital HSV infection in this population may follow a prolonged and severe clinical course. Systemic symptoms may be prominent, and chronic lesions may develop if immunological clearance is impaired. In the absence of ART, primary genital herpes can be severe and prolonged, with progressive, multifocal, and coalescing mucocutaneous anogenital lesions. Serious complications—including meningitis, fulminant hepatitis, pneumonia, neurological disease, and disseminated infection—have been reported.

In the absence of definitive data, most experts recommend higher-than-standard antiviral dosing for initial HSV episodes in immunocompromised individuals. Some clinicians advocate a 10-day course at twice the standard dose of oral antivirals.

However, such intensified regimens may not be necessary for all people living with HIV, particularly those with normal CD4 counts.

For patients with advanced HIV disease: - Double the standard dose of antiviral therapy should be considered. - If new lesions continue to appear between Days 3 and 5 of treatment, the dose should be further increased. - Early initiation of therapy is critical. - If new lesions persist after 3–5 days, repeat viral isolation should be attempted, and susceptibility testing performed if possible. - The antiviral dose should be escalated accordingly.

Recommended Initial Antiviral Doses for All People Living with HIV:

- **Aciclovir:** 400 mg five times daily for 7–10 days
- **Valaciclovir:** 500 mg to 1 g twice daily for 10 days
- **Famciclovir:** 250–500 mg three times daily for 10 days

Treatment should last at least 10 days or until complete re-epithelialization of lesions—often longer than the standard duration used in HIV-negative individuals.

Management of Fulminant Disease:

If severe or disseminated disease occurs: - Switch to intravenous aciclovir at 5–10 mg/kg every 8 hours for 2–7 days, or until clinical improvement. - Follow with oral antiviral therapy to complete a minimum total of 10 days of treatment.

Management of Recurrent HSV Disease in People Living with HIV

Both symptomatic and asymptomatic HSV reactivations are more frequent in people living with HIV. These recurrences may lead to persistent and progressive anogenital mucocutaneous lesions, especially when CD4 counts are below 50 cells/mm³. Lesions may be atypical, larger, deeper, and hypertrophic compared to those seen in immunocompetent individuals.